Nginx and AWS Application Load Balancer (ALB)

1. Introduction

In modern cloud-based architectures, applications must be:

- Scalable
- Highly available
- Secure
- Efficient in handling traffic

To achieve this, technologies like **Nginx** and **AWS Application Load Balancer (ALB)** are used together.

This document explains all critical concepts related to:

- Nginx (web server + reverse proxy)
- ALB (managed load balancer)
- Traffic flow and architecture
- SSL, DNS, and routing concepts

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2. Core Architecture Overview

Request Flow

```
User → Route 53 →
   → Nginx EC2 (Site 1, Site 2)
   → ALB → EC2 Instances (Site 3, Site 4)
```

Key Components

Component	Role
Route 53	DNS routing
Nginx	Web server
ALB	Load balancing
EC2	Application hosting
ACM	SSL certificates

3. Nginx (Deep Explanation)

🔹 3.1 What is Nginx?

Nginx is an open-source web server that can also act as:

- Reverse proxy
- Load balancer
- HTTP cache

It uses an **event-driven architecture**, allowing it to handle thousands of requests efficiently with low memory usage

⚡ 3.2 Core Features

- High performance (handles 10k+ connections)
- Reverse proxy support
- Static file serving
- SSL/TLS support
- URL routing and rewriting
- Virtual hosting

3.3 Important Concepts in Nginx

🔸 a) Web Server

Serves static content like:

- HTML
- CSS
- Images

🔸 b) Reverse Proxy

Nginx forwards requests to backend servers.

Example:

```
User → Nginx → Flask App
```

🔸 **c) Virtual Hosts (Server Blocks)**

Allows multiple websites on a single server.

```
server {
    listen 80;
    server_name site1.domain.com;
    root /var/www/site1;
}
```

🔸 **d) Event-Driven Architecture**

- Handles multiple requests asynchronously
- Uses less memory compared to thread-based servers

🔸 **e) Caching**

Nginx can cache responses → improves performance
(ALB cannot do this natively)

🔸 **f) Load Balancing (Basic)**

Nginx supports:

- Round Robin
- Least Connections

3.4 Role of Nginx in Your Project

- Hosts **Site 1 and Site 2**
- Uses **server blocks**
- Serves static content

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4. AWS Application Load Balancer (ALB)

🔹 **4.1 What is ALB?**

ALB is a **Layer 7 (Application Layer) load balancer** that distributes HTTP/HTTPS traffic across multiple servers

4.2 Core Features

- Load balancing across EC2 instances
- Health checks
- Auto scaling integration
- HTTPS support
- Path-based routing
- Host-based routing

4.3 How ALB Works

1. Receives request from user
2. Evaluates listener rules
3. Forwards request to target group
4. Sends response back

👉 ALB routes traffic only to **healthy targets**

4.4 Important ALB Components

🔸 **a) Listener**

- Listens on ports (80 / 443)
- Defines rules

🔸 **b) Target Group**

- Group of EC2 instances

🔸 **c) Health Check**

- Checks if instance is alive
- Stops sending traffic if unhealthy

🔸 **d) Availability Zones**

- ALB runs across multiple AZs
- Ensures high availability

🔸 **e) Cross-Zone Load Balancing**

- Distributes traffic evenly across all instances

4.5 Role of ALB in Your Project

- Handles **Site 3 and Site 4**
- Distributes traffic between EC2 instances
- Provides scalability and fault tolerance

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5. SSL / HTTPS (ACM)

🔹 5.1 What is SSL?

SSL encrypts communication between user and server.

🔹 5.2 AWS ACM

- Provides SSL certificates
- Used with ALB
- Supports HTTPS

5.3 HTTPS Flow

```
User → HTTPS → ALB → HTTP → EC2
```

SSL termination happens at ALB

5.4 HTTP → HTTPS Redirect

Ensures:

- Secure communication
- No unencrypted traffic

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6. DNS (Route 53)

🔹 6.1 What is DNS?

DNS maps domain names to IP addresses.

🔹 6.2 Route 53 Role

- Routes traffic to:
  - EC2 (Nginx)
  - ALB

Example Mapping

Domain	Target
site1.domain.com	Nginx EC2
site3.domain.com	ALB

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7. Load Balancing Concept

🔹 7.1 What is Load Balancing?

Distributes traffic across servers to:

- Improve performance
- Reduce downtime

🔹 7.2 Why Important?

- Prevents overload
- Improves scalability
- Ensures high availability

Used heavily in web applications handling large traffic

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8. Nginx vs ALB (Deep Comparison)

Feature	Nginx	ALB
Type	Software	Managed AWS service
Layer	L7 (Application)	L7 (Application)
Deployment	EC2	AWS-managed
Scaling	Manual	Automatic
Caching	Yes	No
Routing	Advanced	Rule-based
Maintenance	Required	Not required

Simple Understanding

Nginx = Server + control
ALB = Managed traffic distributor

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9. Why Both Are Used Together

Combined Benefits

- Nginx → efficient content serving
- ALB → scalable traffic distribution

Real-world Insight

- ALB handles **external traffic**
- Nginx handles **internal request processing**

Common architecture in production systems

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10. Advanced Concepts

🔹 **Reverse Proxy vs Load Balancer**

- Reverse proxy → forwards requests
- Load balancer → distributes traffic

🔹 **High Availability**

- ALB works across AZs
- Ensures uptime

🔹 **Scalability**

- ALB auto scales with traffic

🔹 **Fault Tolerance**

- Unhealthy instances removed automatically

🔹 **Single Entry Point**

- ALB acts as entry point for system

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11. Conclusion

- Nginx is used for hosting and processing requests
- ALB is used for distributing traffic
- Together they create a **scalable, secure, and high-performance system**

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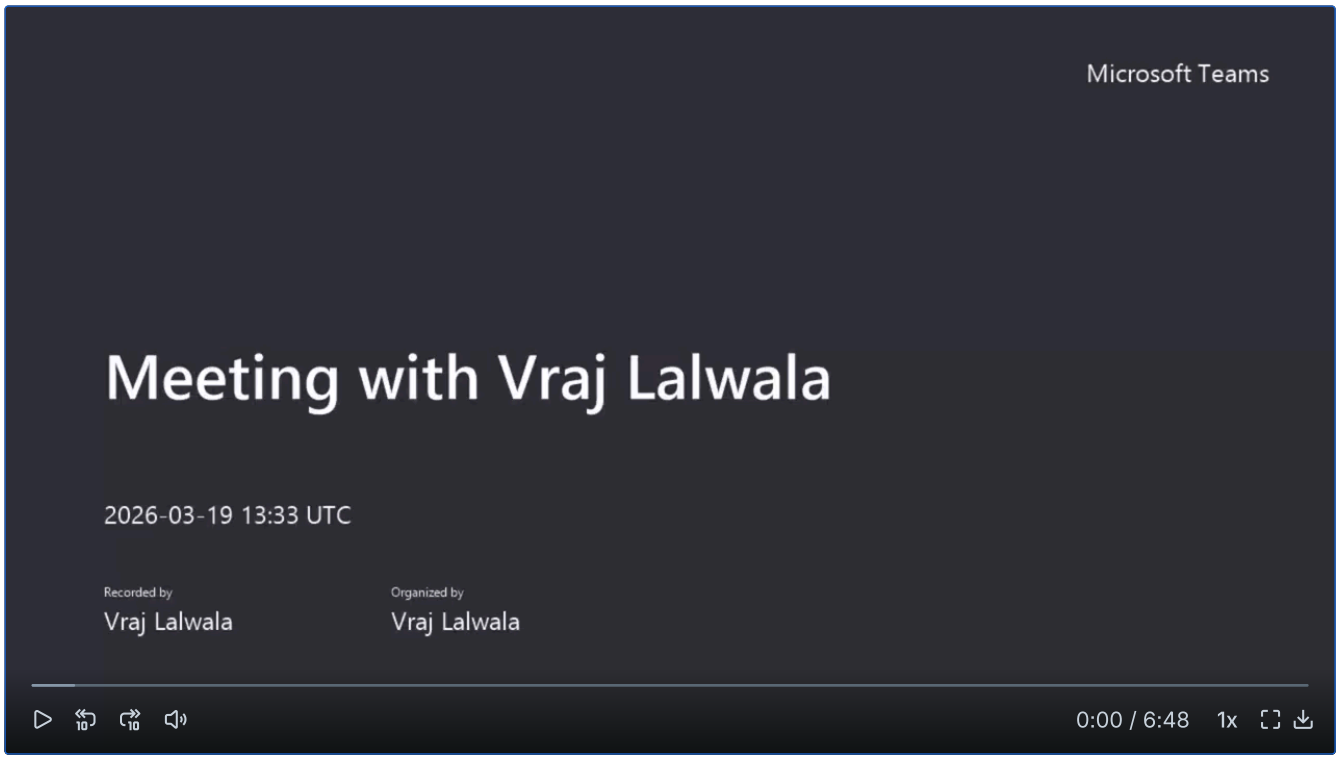
12. Video Explanation